

Lasso and Ridge Regression in R

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Outline

1:00-1:30 Concepts

- Ames Housing Data
- Bias-Variance Tradeoff
- Lasso

1:30-2:15 Live Coding

2:15-2:45 Break ☺

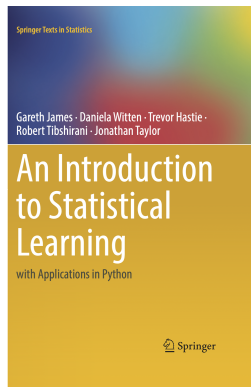
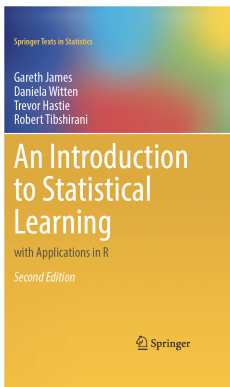
2:45-3:15 Concepts

- Ridge Regression
- Validation

3:15-4:00 Coding Exercises

Questions are **highly** encouraged!

An Introduction to Statistical Learning

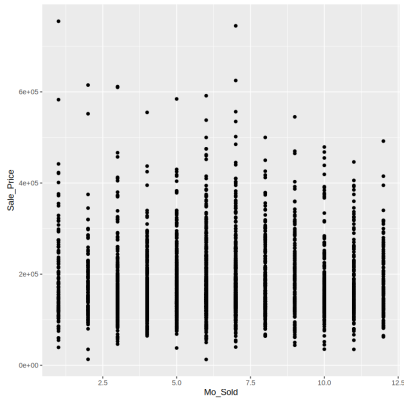
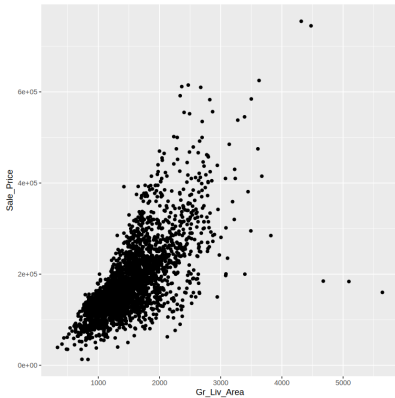


- Available for FREE online: statlearning.com
- Today we'll cover mostly Section 6.2 (in R)

Ames Housing Data

2,930 houses sold between 2006-2010 in Ames, IA

Goal: Predict sale price given 80(!) predictor variables



Forward and Backward Stepwise Selection

Forward Stepwise Selection:

- Start with **none** of the predictors
- At each step, add the single best predictor
- Stop when no additional predictor would be significant

Backward Stepwise Selection:

- Start with **all** of the predictors
- At each step, drop the least significant predictor
- Stop when all remaining predictors are significant

Greedy Algorithms



Credit: Villains Wiki

- Computationally cheap 😊
- Not guaranteed to find the optimal solution ☹️

Best Subset Selection

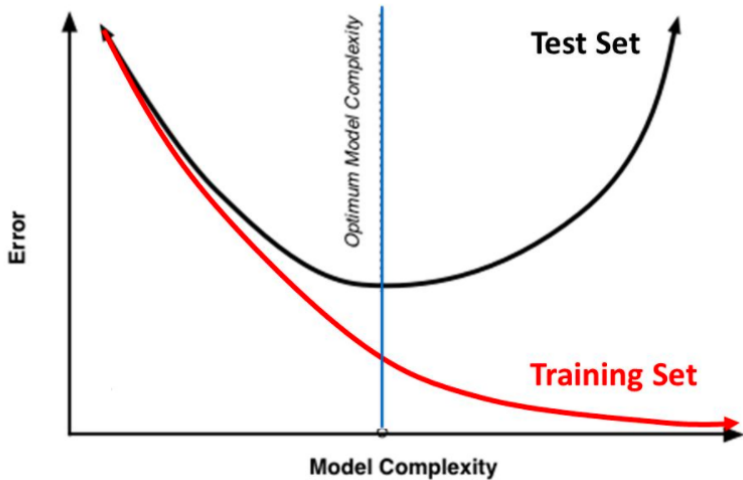
$$Y \approx \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p$$

$$\hat{\beta} = \arg \min_{\beta} \left\{ \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 \right\}$$

subject to $\sum_{j=1}^p \mathbb{I}\{\beta_j \neq 0\} \leq s$

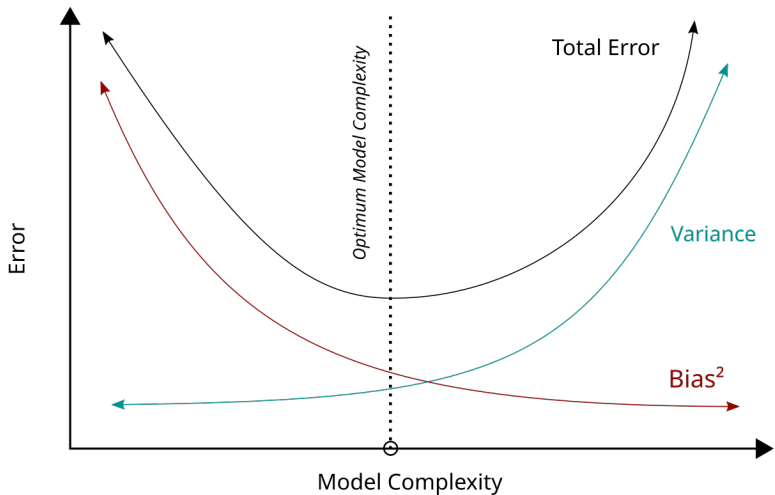
- Computationally expensive
- What happens to the error of the model as s increases?

Training Error vs. Test Error



Credit: Mikhail Dozmorov

Bias-Variance Tradeoff



Credit: Wikipedia

Convex Optimization

$$\hat{\beta} = \arg \min_{\beta} \left\{ \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 \right\}$$

subject to $\sum_{j=1}^p \mathbb{I}\{\beta_j \neq 0\} \leq s \quad \leftarrow \text{hard } \odot$

$$\hat{\beta} = \arg \min_{\beta} \left\{ \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 \right\}$$

subject to $\sum_{j=1}^p |\beta_j| \leq s \quad \leftarrow \text{easy } \odot$

The Lasso

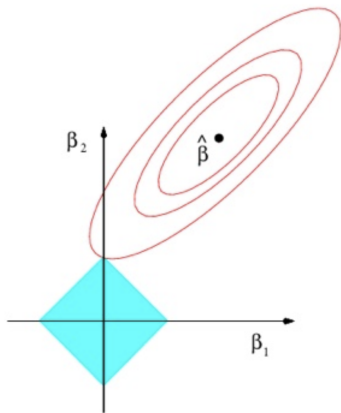
$$\hat{\beta} = \arg \min_{\beta} \left\{ \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 \right\}$$

subject to $\sum_{j=1}^p |\beta_j| \leq s$

Or, equivalently:

$$\hat{\beta} = \arg \min_{\beta} \left\{ \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j| \right\}$$

How does the lasso work?



Credit: ISLR, page 245

Exercise (ISLR, page 284)

The lasso, relative to least squares, is: (choose correct answer)

- i. More flexible and hence will give improved prediction accuracy when its increase in bias is less than its decrease in variance.
- ii. More flexible and hence will give improved prediction accuracy when its increase in variance is less than its decrease in bias.
- iii. Less flexible and hence will give improved prediction accuracy when its increase in bias is less than its decrease in variance.
- iv. Less flexible and hence will give improved prediction accuracy when its increase in variance is less than its decrease in bias.

Live Coding

Ground Rules

- After opening the Google Colab notebook, select “File” → “Save a Copy in Drive”. This will open a new tab with a version you can edit and save. If you keep the original open in a separate tab, you can refresh it to see updates I post.
- Next, select “Runtime” → “Change runtime type”, and change the runtime type from Python 3 to R.

Exercises

1. Load, clean and visualize the Ames Housing data. Split the data into a training set and a test set.
2. Try to fit a best subset selection model on the training set to predict $\log(\text{Sale Price})$. What happens?
3. Fit a lasso regression model on the training set to predict $\log(\text{Sale Price})$, and interpret the results.

Break 😊

Ridge Regression

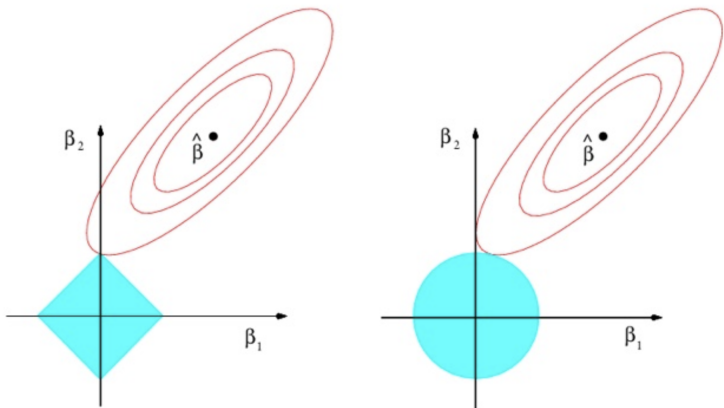
Lasso:

$$\hat{\beta} = \arg \min_{\beta} \left\{ \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j| \right\}$$

Ridge Regression:

$$\hat{\beta} = \arg \min_{\beta} \left\{ \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right\}$$

How does the ridge regression work?



Credit: ISLR, page 245

Exercise (ISLR, page 284)

Suppose we estimate the coefficients in a linear regression model by minimizing

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2$$

- (a) As we increase λ from 0, the training RSS will: (choose correct answer)
- i. Decrease initially, and then eventually increase in a U shape.
 - ii. Increase initially, and then eventually decrease in an inverted U.
 - iii. Steadily increase.
 - iv. Steadily decrease.
 - v. Remain constant.
- (b) Repeat (a) for test RSS.
- (c) Repeat (a) for variance.
- (d) Repeat (a) for (squared) bias.

Elastic Net Regression

$$\sum_{i=1}^n w_i \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + (1 - \alpha) \cdot \lambda \sum_{j=1}^p \beta_j^2 + \alpha \cdot \lambda \sum_{j=1}^p |\beta_j|$$

```
glmnet::glmnet(  
  x,  
  y,  
  weights = NULL,  
  alpha = 1,  
  lambda = NULL,  
  standardize = TRUE,  
  intercept = TRUE,  
  ...  
)
```

```
glmnet::cv.glmnet(  
  x,  
  y,  
  weights = NULL,  
  lambda = NULL,  
  nfolds = 10,  
  foldid = NULL,  
  parallel = FALSE,  
  ...  
)
```

Validation

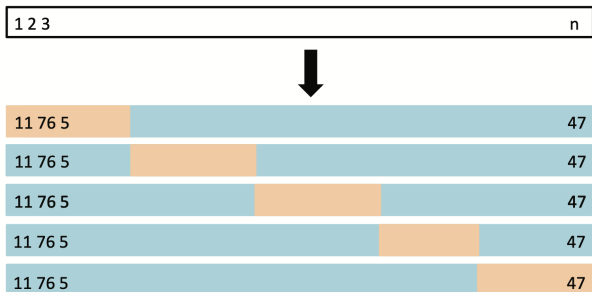
How to select the tuning parameter λ ?

Validation set approach:



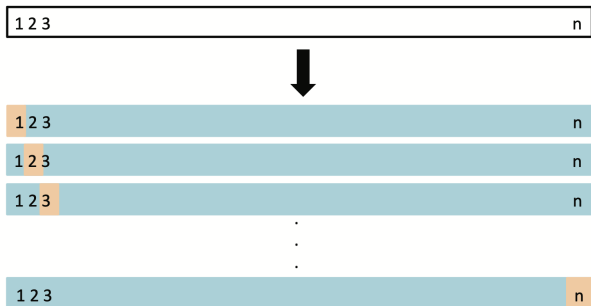
Credit: ISLR, page 199

k -Fold Cross-Validation



Credit: ISLR, page 203

Leave-One-Out Cross-Validation (LOOCV)



Credit: ISLR, page 201

Exercise (ISLR, page 220)

- (a) Explain how k -fold cross-validation is implemented.
- (b) What are the advantages and disadvantages of k -fold cross-validation relative to:
 - i. The validation set approach?
 - ii. Leave-one-out cross-validation (LOOCV)?
- (c) What is the trade-off to consider when choosing k for k -fold cross-validation?

Coding Exercises

Exercises

1. Fit a ridge regression model on the training set to predict $\log(\text{Sale Price})$.
2. Plot the ridge coefficients against the lasso coefficients, and interpret the results.
3. Compare the test error of ridge regression vs. lasso.

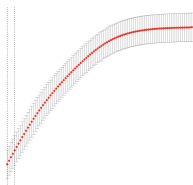
Bonus (if you have extra time)

4. Try fitting forward and backward stepwise selection models on the training set to predict $\log(\text{Sale Price})$. Compare the test errors of these models against lasso and ridge regression.
Hint: See Section 6.5 of ISLR (statlearning.com)

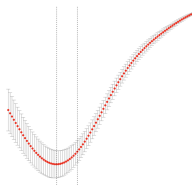
Three glmnet lessons I learned the hard way

```
glmnet::glmnet(  
  x,  
  y,  
  weights = NULL,  
  alpha = 1,  
  lambda = NULL,  
  standardize = TRUE,  
  intercept = TRUE,  
  ...  
)
```

1. For ridge regression, use $\alpha = 0$.
2. Check whether the default `lambda` range includes the optimal value for you



Example of bad λ range



Example of good λ range

3. For RAPM, `standardize = FALSE`.